

Jimma Institute of Technology
B.Sc. Materials Science and Engineering
Tutorial - Multiple Choice Questions

Model Exam – 4

12.06.2023

Name of the student	Register number

Answer the following questions.

1. Which of the following polymer processing methods is commonly used to produce plastic bottles?
(a). Extrusion (b). Injection molding (c). Thermoforming (d). Blow molding
Answer: d) Blow molding
2. Which polymer processing method involves the simultaneous extrusion of multiple polymers?
(a). Compression molding (b). Coextrusion (c). Transfer molding (d). Rotational molding
Answer: b) Coextrusion
3. What is the primary purpose of injection molding?
(a). Producing fibers and filaments (b). Creating three-dimensional shapes (c). Coating surfaces with a thin polymer layer (d). Blowing air into a polymer tube
Answer: b) Creating three-dimensional shapes
4. Which of the following polymer processing methods is commonly used to produce plastic films and sheets?
(a). Blow molding (b). Calendering (c). Pultrusion (d). Dip coating
Answer: b) Calendering
5. What is the primary advantage of rotational molding?
(a). High production speed (b). Precise control of part dimensions (c). Ability to create hollow parts with complex shapes (d). Low cost of equipment
Answer: c) Ability to create hollow parts with complex shapes
6. Which polymer processing method involves pouring liquid polymer into a mold and allowing it to solidify?
(a). Extrusion (b). Injection molding (c). Polymer casting (d). Film blowing
Answer: c) Polymer casting
7. What is the primary purpose of thermoforming?
(a). Creating fibers and filaments (b). Producing thin polymer films (c). Shaping thermoplastic sheets into three-dimensional objects (d). Coating surfaces with a polymer
Answer: c) Shaping thermoplastic sheets into three-dimensional objects
8. Which polymer processing method uses a combination of heat and pressure to shape a polymer between two mold halves?
(a). Compression molding (b). Transfer molding (c). Injection molding (d). Calendering

Answer: a) Compression molding

9. What is the primary purpose of film blowing?

- (a). Producing plastic bottles (b). Creating fibers and filaments (c). Manufacturing plastic films and sheets (d). Shaping thermoplastic sheets into three-dimensional objects

Answer: c) Manufacturing plastic films and sheets

10. Which polymer processing method is commonly used to produce rubber products?

- (a). Extrusion (b). Blow molding (c). Injection molding (d). Reaction injection molding

Answer: a) Extrusion

11. What is the primary advantage of extrusion?

- (a). Ability to create complex shapes (b). High production speed
(c). Precise control of part dimensions (d). Low cost of equipment

Answer: b) High production speed

12. Which polymer processing method involves the use of molds and a chemical reaction to produce complex parts?

- (a). Thermoforming (b). Reaction injection molding (c). Coextrusion (d). Pultrusion

Answer: b) Reaction injection molding

13. What is the primary purpose of blow molding?

- (a). Producing fibers and filaments (b). Creating three-dimensional shapes (c). Shaping thermoplastic sheets into objects (d). Coating surfaces with a thin polymer layer

Answer: b) Creating three-dimensional shapes

14. Which polymer processing method involves heating and pressurizing a thermosetting resin in a closed mold?

- (a). Transfer molding (b). Compression molding (c). Extrusion (d). Calendering

Answer: a) Transfer molding

15. What is the primary advantage of injection molding?

- (a). Ability to produce complex shapes with high precision (b). High production speed (c). Low cost of equipment (d). Suitable for producing large objects

Answer: a) Ability to produce complex shapes with high precision

16. Which polymer processing method is commonly used to produce composite materials with continuous fiber reinforcement?

- (a). Injection molding (b). Compression molding (c). Pultrusion (d). Thermoforming

Answer: c) Pultrusion

17. What is the primary purpose of calendering?

- (a). Creating fibers and filaments (b). Producing plastic bottles (c). Manufacturing plastic films and sheets (d). Shaping thermoplastic sheets into objects

Answer: c) Manufacturing plastic films and sheets

18. Which polymer processing method involves the use of a heated die to shape a polymer by forcing it through a small opening?

- (a). Extrusion (b). Injection molding (c). Blow molding (d). Polymer casting

Answer: a) Extrusion

19. What is the primary advantage of compression molding?

(a). High production speed (b). Precise control of part dimensions (c). Ability to create complex shapes (d). Suitable for producing large objects

Answer: c) Ability to create complex shapes

20. Which polymer processing method is commonly used to create thin coatings on surfaces?

(a). Thermoforming (b). Dip coating (c). Reaction injection molding (d). Film blowing

Answer: b) Dip coating